Turn legal expertise into reliable software.

Learn the decisions.
Make the rules explicit.
Check the result with the team.

The lawyer decides legal applicability.

LAWYERS

Approve the reasoning

RULES

Assemble and explain

AI REVIEW

Suggest checks and sources

Basis: original internship posting supplied by the candidate. Proposed understanding, not an internal-system assessment.

My earlier work in legal question answering

North South University · Bangladeshi statutes · early research contribution

1

Review the problem

Literature review and problem identification

2

Organise sources

Initial design: keep sections and their Act names together

3

Move to LLM work

Another sub-team continued the legal research

Earlier research asked: “Where is the evidence?”
This recent demo also asks: “When should the paragraph be used?”

Earlier responsibilities based on candidate account. The recent prototype is separate from the published implementation.

Define the sources, paragraphs and their rules

Cell 1 stores the evidence. Cell 2 says when each prepared paragraph can be used.

SOURCE RECORD · C167

Act + section number
Prepared summary text
Status + recorded judgment note

10 labelled sample sources

→

PARAGRAPH RULE · M5 RECORDS

Condition	Records available = True
Dependency	Procedural block M3 included
Order / citation	Position 50 / source C167
Text	A prepared records paragraph

Output: 10 source records + 7 module definitions, held in knowledge-base version 1.2.0.

Notebook cells 1–2: 10 educational source summaries; 7 modules; fictional judgment updates and teaching rules.

Show exactly why each paragraph is selected

Validate facts → check conditions and dependencies → reject conflicts → fill templates.

SAMPLE INPUT

Forum: magistrate
Facts: available
Procedure: requested
Records: available
Relief: requested
Other route flags: false

BLOCK	DECISION	WHY
M1	**INCLUDED**	Forum matches
M2	**INCLUDED**	Facts available + M1 included
M3	**INCLUDED**	Procedure requested + M2 included
M4	**SKIPPED**	Substantive route flag is false
M5	**INCLUDED**	Records available + M3 included
M6	**SKIPPED**	Bail flag is false
M7	**INCLUDED**	Relief requested + M2 included

Output: Jurisdiction → Facts → Procedure → Records → Relief, plus this decision trace.

Notebook cell 3: actual selected sequence for fictional SAMPLE_CASE; no generative model assembles this draft.

How search and query decomposition happen

Controlled example input: C54 section text ; P420 section text. Threshold tau = 0.80.

1 · SEARCH WHOLE QUERY

Best score: 0.7879

Below 0.80 → no accepted hit

→

2 · SPLIT THE ISSUES

Split at the semicolon

C54 text / P420 text

→

3 · SEARCH EACH PART

Exact-text matches ≈ 1.0

Both pass the same gate

→

4 · MERGE RESULTS

Return C54 + P420

Remove duplicate sources

SEPARATE DRAFT REVIEW

Citation markers + recorded dates → M3/C54, M3/C167 and M5/C167 receive update warnings.

The ordinary case summary has no accepted sources. Exact-text success is not a retrieval-quality benchmark.

What each regression test is trying to catch

Each test starts with an input or change, then checks a written expectation.

TEST	INPUT / CHANGE	WHAT WE EXPECT
Repeatability	Same case run 25 times	No drift in text or full result
Reference-case parity	Four fixed synthetic cases	Expected blocks, order and citations
Conflict guard	Turn both exclusive routes on	Reject them with a consistent error
Update alerting	Add a simulated C167 judgment note	Flag M3 and M5 as needing review

All four groups PASS. Extra assertions check missing/unknown citations and retrieval paths.

Notebook cell 5: four test groups passed; fixtures are synthetic, not genuinely lawyer-approved historical files.

Two practical questions behind the implementation

What should count as a match—and how can a change be checked?

1 · Which search results should be accepted?

Metric: cosine similarity on TF-IDF vectors.
Threshold: accept a score of 0.80 or higher.

Why: a starting value from the reference approach.
Meaning: a similarity gate, not 80% accuracy.

2 · How can an unintended output change be detected?

Choice: fixed expected modules and citations;
SHA-256 fingerprints for repeated outputs.

Why: do not rely on rereading each draft alone.
Meaning: detect differences, not prove legal correctness.

Next: calibrate retrieval on realistic questions and validate expectations with lawyers.

Design rationale for this demonstration. 0.80 is uncalibrated; reference cases and updates are synthetic.

PROPOSED NEXT STEPS

How I would keep it reliable in real use

Start with one approved workflow, then expand only after review.

BEFORE THE PILOT	**Use approved material**	Verified full sources, agreed rules and authorised example cases.
WITH EACH CHANGE	**Version, test and review**	Rerun agreed cases; explain intended and unexpected differences.
BEFORE EXPANDING	**Evaluate the weak points**	Measure retrieval and citation support; keep legal decisions with reviewers.

Which workflow would be the most useful starting point for this internship?

Proposed improvements based on the brief and demo limitations, not verified shortcomings in the existing company system.